LendingClub Loan Recovery Ratio Prediction

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Motivation

Peer-to-Peer lending platforms like LendingClub became popular after the Great Recession when credit tightened. There is growing interest in private credit because investors seek diversified, higher-yielding investments and because US household debt is at a record 18.59 trillion.

There is more utility in predicting the recovery ratio. We want to learn from LendingClub's data to help inform investment decisions in a future credit-tightening cycle.

Question to Answer: **What is an expected recovery ratio of a certain loan?**
Loan Recovery Ratio = Total Principal Payment / Total Loan Amount

Related Work

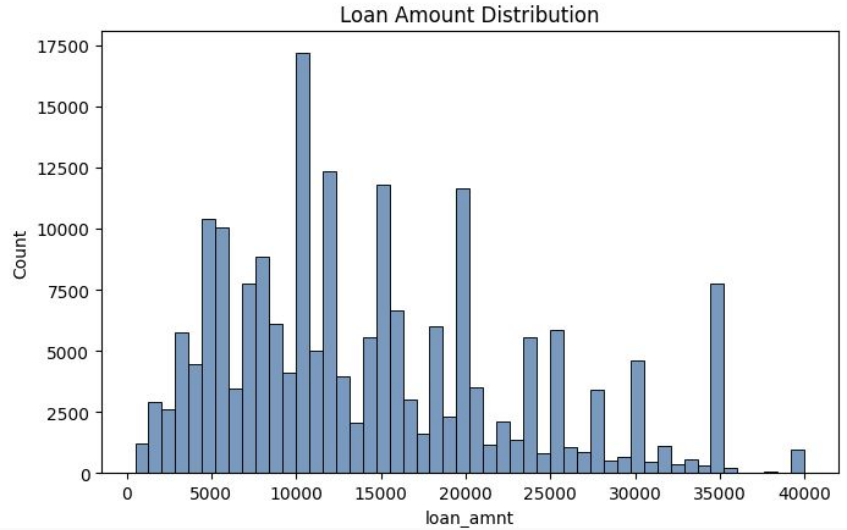
- Trinh 2024: used a similar peer-to-peer lending data and performed a binary **classification** (default vs. fully paid). ➡ Gradient Boosting Decision Tree is the best model.
- Change 2015: used LendingClub data and fit Logistic Regression, SVM, and Naive Bayes for loan default **classification** (default vs non-default). ➡ Naive Bayes with Gaussian performs the best.
- Souadda 2025: evaluated hyperparameter optimization methods for machine learning models with similar lending data for loan default **classification**. ➡ Bayesian methods (Hyperopt, Optuna) achieved 75.7 times faster runtime than Grid Search with similar accuracy.

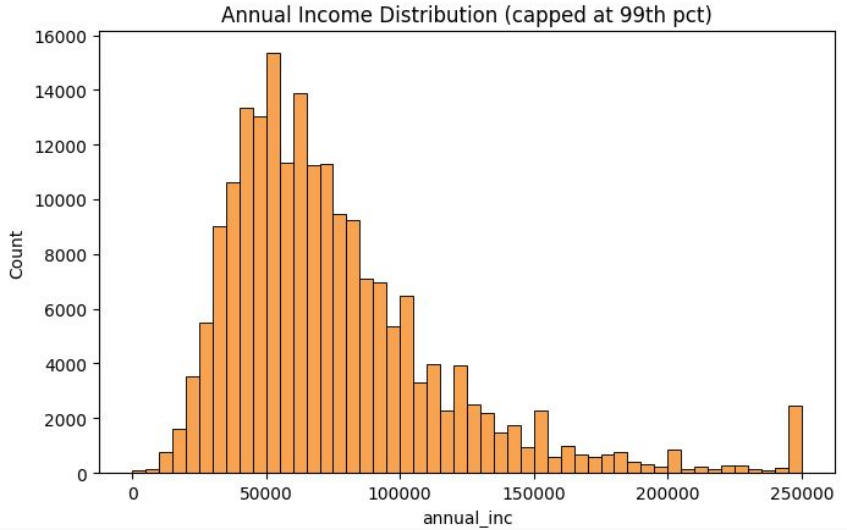
- Existing work focuses on a binary prediction of loan default. Our work **predicts loan recovery ratios** based on each loan's characteristics, which is **unique** among existing literature.

Data

- Kaggle dataset derived from LendingClub peer-to-peer lending data.
- The dataset covers all loans originated between 2007 and 2015 and includes borrower information, loan characteristics, and final loan outcomes.
- The raw data contains 2,260,668 observations with 145 features. We are only keeping finalized loans—those with a `"Default"`, `"Charged Off"`, or `"Fully Paid"` status.
- Categorical and numerical features are processed separately using a preprocessing pipeline. Numerical variables (23 in total) are imputed using the median and standardized, while categorical variables (6 in total) are imputed using the most frequent category and encoded via one-hot encoding.
- We are splitting the data by time to avoid leakage. The training data consists of 912,546 observations in 8 input features. The validation data, which follows the training data chronologically, consists of 199,546 observations. The test data—last chronologically—consists of 199,546 observations as well.

Exploratory Data Analysis

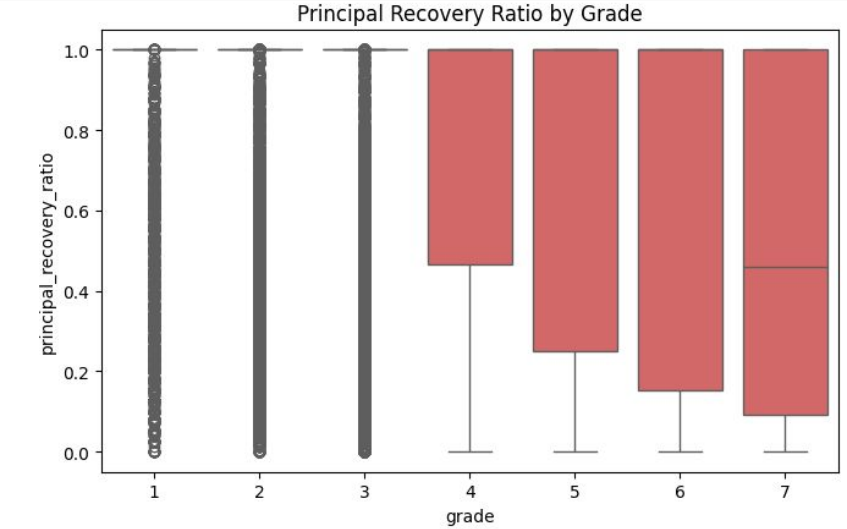


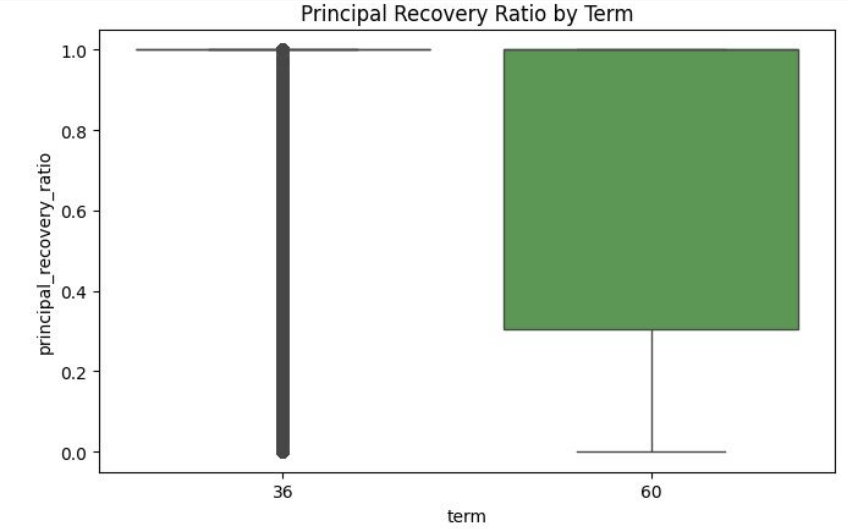


The distribution of loan amounts is not normal and shows multiple peaks, which may reflect common loan size bands or product constraints. There is also a moderate right tail, with fewer observations at larger loan amounts.

Annual income is strongly right-skewed. Most observations are concentrated in the lower-to-middle income range, with a long tail toward higher incomes. Even after capping at the 99th percentile, the distribution still shows substantial positive skew.

Exploratory Data Analysis

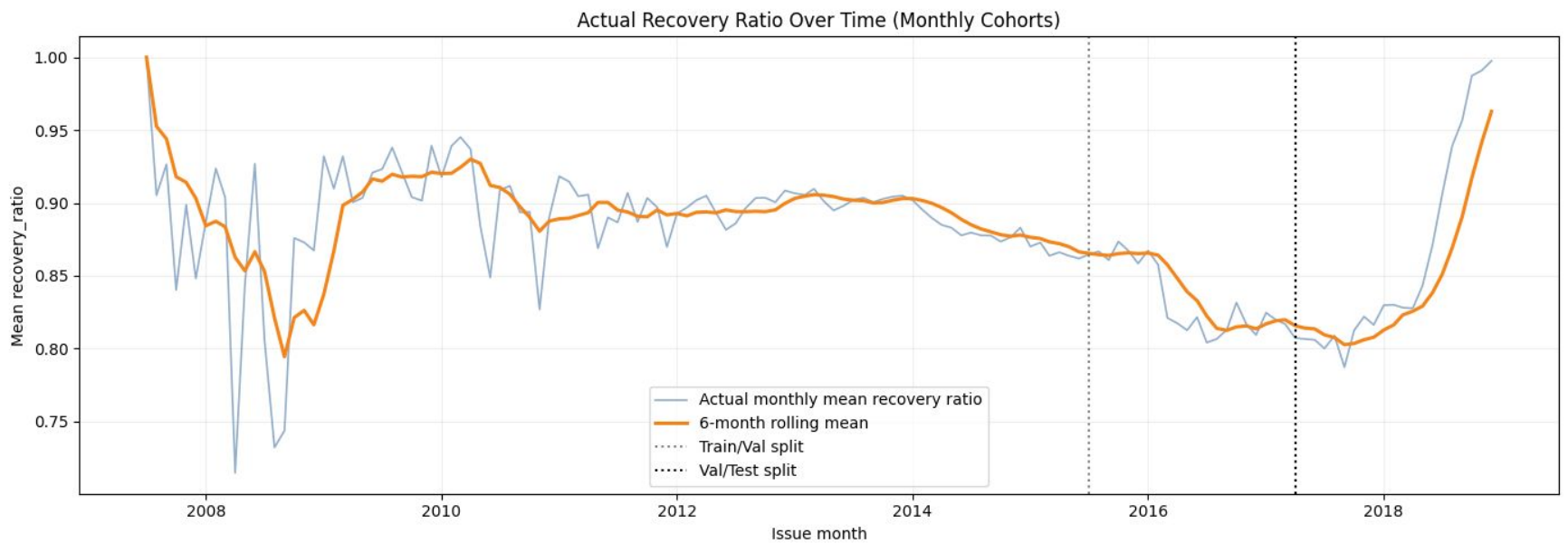




Borrowers with worse grades tend to recover less principal, which is consistent with higher credit risk.

Longer-term loans seem riskier from a recovery perspective, since they are more likely to end up with lower principal recovery.

Exploratory Data Analysis



Loan Recovery Ratios differ across time span in different periods.

Models

Baseline: Linear Regression

A linear regression of principal recovery ratio against the 101 input features.

Enhanced Model 1: XGBoost

The model operates on tabular feature vectors with no sequential structure. The model consists of an ensemble of 700 decision trees trained sequentially using gradient boosting.

Enhanced Model 2: Long Short-Term Memory

The model was adapted to loan-level observations by representing each sample at a single timestep with multiple features.

Models - Hyperparameters

XGBoost

- n_estimators = 700: appropriate number of learners to reduce bias; paired with a moderate learning rate for faster converge without overfitting.
- max_depth = 4: Shallower trees to control overfitting on noisy loan-level outcomes.
- learning_rate = 0.04: Given large number of trees, larger learning rate was picked to balance out training speed and better generalization
- subsample = 0.85, colsample_bytree = 0.9: to improve robustness and reduce variance
- reg_lambda (L2) : L2 regularization to reduce overfitting
- objective = squared error: Standard regression loss, aligning with baseline for comparison

Tuning Process: Finding a balanced setting with appropriate complexity, without overfitting. Grid search performed to find the ebst learning rate.

Models - Hyperparameters

LSTM

- Sequence length = 1 timestep (features as a single step): Matches one loan = one row
- 32 LSTM hidden units with 0.2 dropout: Small capacity relative to sample size and feature dimension; reduces overfitting vs a wide network.
- Dense(16) + ReLU then 0.2 dropout: Lightweight head maps LSTM state to a scalar; second dropout regularizes the mapping to the output.
- Single linear output: Direct regression to recovery ratio.
- Optimizer: Adam: Adaptive steps; works well with default-ish settings on this scale.
- MAE as loss metrics: More robust than MSE to heavy-tailed / skewed errors in recovery ratio.
- Early stopping on validation MAE: Stops when generalization stops improving to avoid overfitting.

Tuning Process: Finding the appropriate numbr of layers to achieve high performance without overfitting, adding dropout and early stopping to avoid overfitting.

Results

Model	Train MAE	Train RMSE	Val MAE	Val RMSE	Test MAE	Test RMSE	R^2
Baseline: Linear Regression	0.2036	0.2784	0.2041	0.2789	0.2042	0.2790	0.1164
XGBoost	0.1956	**0.2733**	0.1968	**0.2747**	0.1970	**0.2750**	0.1420
LSTM	**0.1209**	0.2883	**0.1823**	0.3599	**0.1780**	0.3758	N/A

MAE as the primary metric: direct interpretation, better answers the prediction accuracy to business partners.
RMSE captures errors from outliers, checks tail behavior and worst-case scenario.
MAPE does not perform well when recovery ratio are very close to 0.

LSTM achieves the best train, validation, test MAE consistently, while XGBoost has better RMSE.
LSTM wins for average error reduction.
XGBoost wins for minimizing errors at extremes.

Conclusions

- XGBoost captures modest nonlinear improvements, and the LSTM achieves the lowest mean absolute error, indicating stronger ability to model complex relationships among borrower and loan characteristics.
- LSTM exhibits higher variance and a systematic positive bias in predictions, while XGBoost offers more stable and better-calibrated outputs.
- While more flexible models improve performance, one should use caution to interpret results based on their business priority and risk tolerance.
- Whether average accuracy outweighs systematic bias should be a careful business tradeoff among stakeholders.

Conclusions

Limitations

- Dataset covers loans from 2007 and 2015 only, limiting the relevance of current environments.
- Model lacks macroeconomic indicators that could influence recovery outcomes.
- The use of an LSTM on effectively non-sequential data may not be natural, leading to instability and bias.
- Evaluation on MAE, RMSE, R^2 do not directly translate to investor-relevant considerations such as portfolio-level returns.

Future Work

- Incorporating time-varying macroeconomic features (unemployment rates or interest rates) to improve model robustness across economic cycles.
- Experimenting with alternative model architectures to improve performance.
- Addressing the bias in predictions through calibration techniques or modified loss functions.
- Extending the analysis to a portfolio optimization framework and better connect the modeling results to investment strategies.

Thank you.

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